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جامعة العلوم التطبيقية الخاصة
كلية تكنولوجيا المعلومات
تقييم امتحان على نهائي 2024-2

APPLIED SCIENCE PRIVATE UNIVERSITY (ASU)

FACULTY OF INFORMATION TECHNOLOGY (FIT)

Final Lab Exam 2024-2

Subject: *Object-Oriented Programming (1) 1301108*

Date: Jun. 4, 2025

Time: 08:40 AM

Student N

Student I

☑ CS ☐ SE ☐ CCC ☐ DSAI ☐ Other

PLEASE NOTE THAT THIS LAB QUIZ IS 3 PAGES AND WITH A TOTAL OF 10 POINTS.

Exam Instructions:

- Use NetBeans IDE to solve the questions below.

Question 1

[10 POINTS]

A. [4.5 marks] Write an abstract class named **AIService**. The class should have:

- Two private attributes: **aiServiceName** (type String) and **aiModel** (type String).
- A constructor to initialize **aiServiceName** and **aiModel** .
- Accessor (**getters**) methods for **aiServiceName** and **aiModel**.
- An **abstract method** named **estimateUsageCost()** to return a ***double*** representing the cost of using the AI service.
- A final method named **getAIServiceSummary()** that returns a String containing a brief summary in the format:

  "***Service: [aiServiceName], Model: [aiModel]***"

Note: **Do not** add **mutators** or a **toString()** method to the **AIService** class.

B. [4.5 marks] Write a class named **ChatbotService** that extends the **AIService** class. The **ChatbotService** class should have:

- A private attribute **numberOfRequests** (type int).
- A private attribute **costPerRequest** (type double, in dollars).
- A constructor to initialize **aiServiceName** , **aiModel**, **numberOfRequests**, and **costPerRequest**.
- An implementation of the **estimateUsageCost()** method using the formula:

  Cost = numberOfRequests × costPerRequest
- A **toString()** method that returns a readable string containing the chatbot service details.

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C. [1 mark] Write a class named **AIServiceDemo** with a main method to:

- Create a ChatbotService object with the following values:
  - 3 ○ **aiServiceName : "ChatBot"**
  - 4 ○ **aiModel: "GPT-4"**
  - 1 ○ **numberOfRequests: 2000**
  - 2 ○ **costPerRequest: 0.002**
- Use both the **estimateUsageCost()** and **toString()** methods to display the cost and summary of the service.

class 1

```
Public abstract class AIService {
private String aiServiceName;
private String aiModel;
public AIService (string aiServiceName, String aiModel) {
        this.aiServiceName = aiServiceName;
        this.aiModel = aiModel ; }
Public String getAiServiseName() {
    return aiServiseName ;
    }
Public string getAiModel () {
        return aiModel ;
    }
Public abstract double estimateUsageCost();
```

3

Class ①

```
public final string getAIServiseSammary() {
return " servise :[aiServiseName] , Model :[aiModel]";
}
}
```

-0.25

Class ②

```
public class chatbotServize extend AIService {
private int numberOfRequests;
private double cost PerRequest;
public chatbotServeice (int numberofRequests, double cost PerRequest
, string aiServeiceNrame , string aiModel) {
    super (aiServiceName , aiModel);
    this.numberOfRequests = numberOfRequests;
    this. cost PerRequest = cost PerRequest;
}
public double estimateUsageCost () {
    return numberofRequests * cost PerRequest;
}
```

class (2)

```
public string toString(){

return "Model : " + super.getAiModel() + " ,name : " +
super.getAiServiceName() + " , number of Requests : " +
numberOfRequests + " , cost Per Request = " + costPerRequests
```